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NN-H-93a

APRIL 2, 1948

SUPERSEDING
Fed. Spec. NN-H-93
July 2, 1941

FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

HANDLES; HICKORY, STRIKING-TOOL

This specification was approved on the above date by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government, and shall become effective not later than July 15, 1948, for use in procurement initiated on or after that date. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS

A–1. There are no other Federal specifications applicable to this specification.

B. TYPES, SIZES, AND GRADES

B–1. *Types and sizes.*—Hickory striking-tool handles, in accordance with this specification, shall be furnished in the following types and sizes, as specified in the invitation for bids:

- Sledge hammer
  - 30-inch
  - 32-inch
  - 36-inch medium
  - 36-inch heavy
- Riveting hammer
  - 12-inch
  - 14-inch
- Single-bitted ax
  - 36-inch
- Double-bitted ax
  - 36-inch
- Claw hammer
  - 12-inch
  - 14-inch
- Adz
  - 34-inch
- Railroad or clay pick
  - 36-inch
- Blacksmith's hammer
  - 14-inch
  - 16-inch
  - 18-inch
- Hatchet
  - 14-inch
  - 16-inch
- Machinist's hammer
  - 10-inch
  - 12-inch
  - 14-inch
  - 16-inch
  - 18-inch

B–1a. *Special types and sizes.*—Types other than those listed above may be purchased under this specification. They shall be of the size and form and for the purpose specified in the invitation for bids and shall otherwise conform to this specification. Sizes other than those listed above may be purchased under this specification, and shall be as specified in the invitation for bids. (See par. E–1a.)

B–2. *Grades.*—Hickory striking-tool handles in accordance with this specification shall be furnished in the following grades, as specified in the invitation for bids:

Grade AA.
Grade A.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—The handles shall be of hickory[1] seasoned to not more than 12 percent moisture content and conforming to the requirements for rate of growth and density shown in table I for the individual grades.

C–2. *Workmanship.*—Workmanship and manufacture shall be in accordance with the best commercial practice. All handles shall be free from defects and blemishes, except as noted in table I.

D. GENERAL REQUIREMENTS

D–1. *Natural color.*—There are no requirements as to the natural color of the wood in the handles, that is, handles of any grade may be heartwood (red hickory), or sapwood (white hickory), or may contain both heartwood and sapwood.

D–2. *Finish.*—The handles shall be smoothly finished by sand and waxing belt operations. The handles shall not be painted.

D–2a. *Stain.*—Handles driven into tool heads by the manufacturers of the heads may be color-stained by the manufacturer provided the staining does not conceal the annual rings and the grain of the wood, does not discolor the hands of the user, and is not objectionable to the Government agency making the purchase.

E. DETAIL REQUIREMENTS

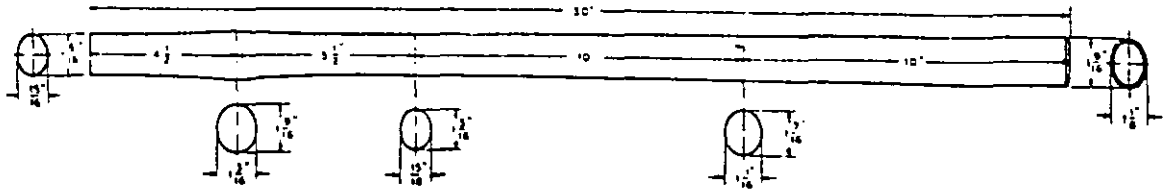
E–1. *Dimensions.*—The handles shall conform to the following drawings with a tolerance of plus or minus ½ inch in over-all length and with a tolerance of plus or minus 1/16 inch in transverse dimensions. However, single-bitted and double-bitted ax handles with eye ends 4½ inches instead of 5 inches long shall be accepted provided they conform to this specification in all other particulars.

E–1a. *Special types or sizes.*—Special types or sizes of handles shall conform to drawings furnished by the Government agency making the purchase, and shall be within the tolerances stated in paragraph E–1.

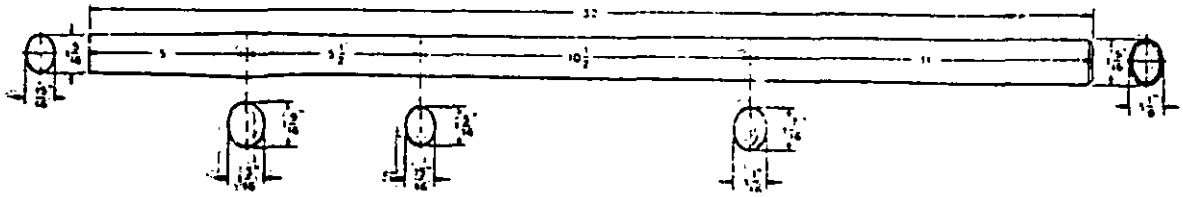
[1] The word "hickory" as used in this specification includes wood cut from the four true hickories of commercial importance: Shagbark hickory (*Carya ovata*, syn. *Hicoria ovata*), shellbark hickory (*Carya laciniosa*, syn. *Hicoria laciniosa*), pignut hickory (*Carya glabra*, syn. *Hicoria glabra*), and mockernut hickory (*Carya tomentosa*, syn. *Hicoria alba*).

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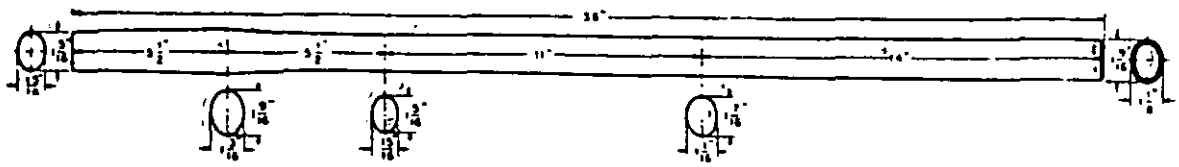
SLEDGE HAMMER HANDLE—30"



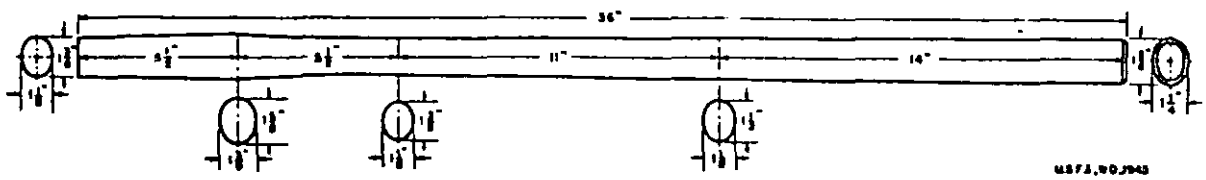
SLEDGE HAMMER HANDLE—32"



SLEDGE HAMMER HANDLE—36" MEDIUM
FOR SLEDGES UP TO AND INCLUDING 16 POUNDS

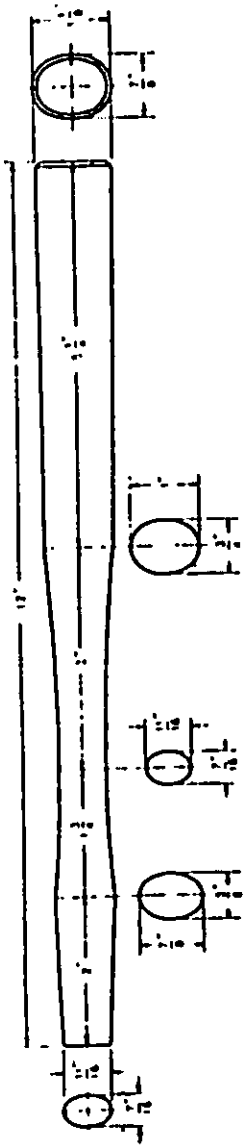


SLEDGE HAMMER HANDLE—36" HEAVY
FOR SLEDGES HEAVIER THAN 16 POUNDS

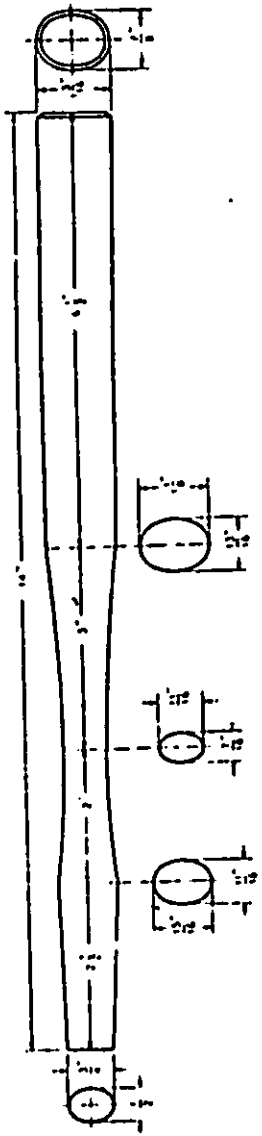


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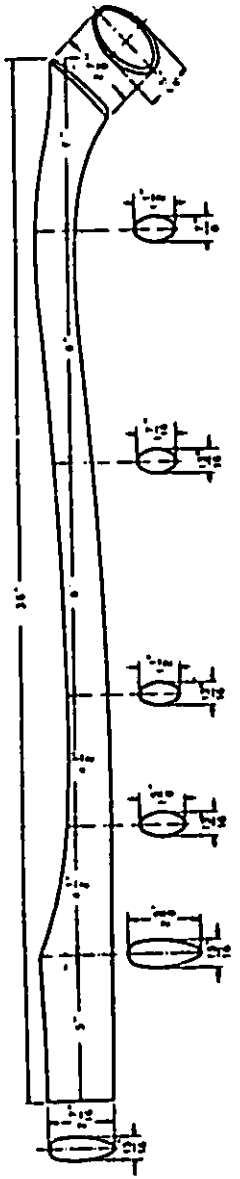
RIVETING HAMMER HANDLE—12"



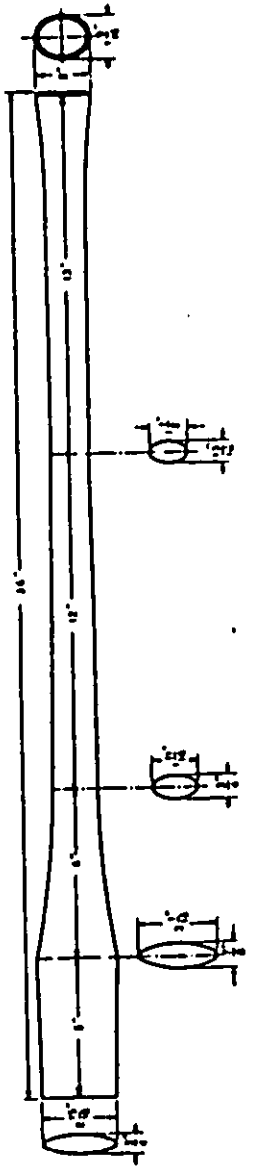
RIVETING HAMMER HANDLE—14"



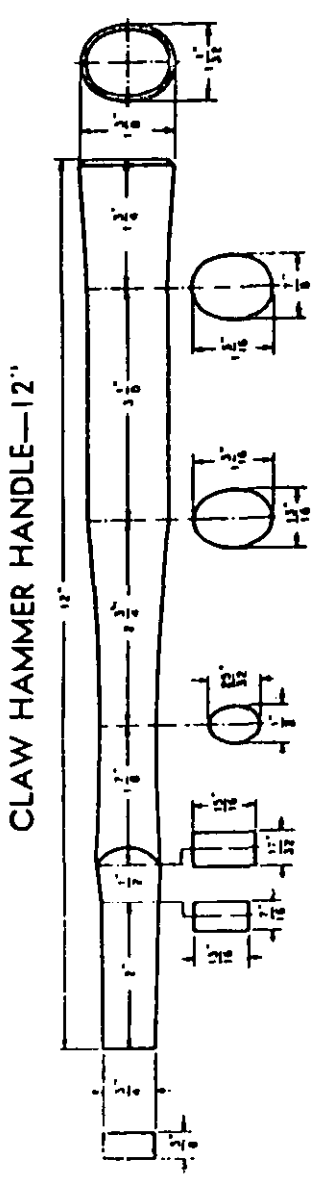
SINGLE-BITTED AX HANDLE—36"

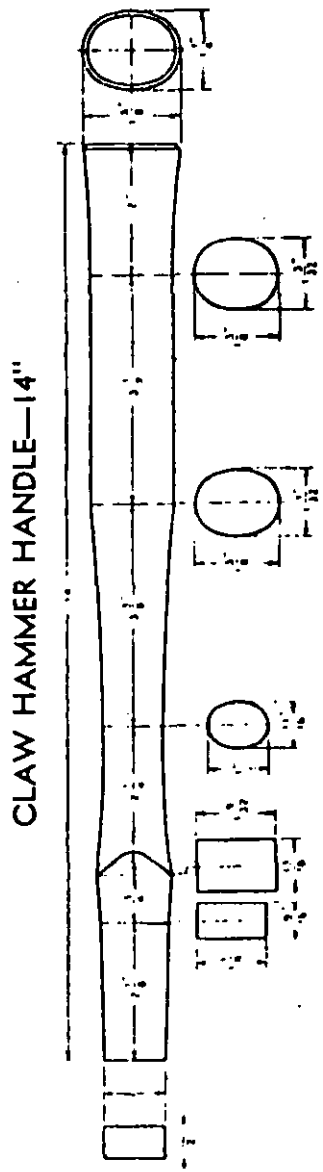


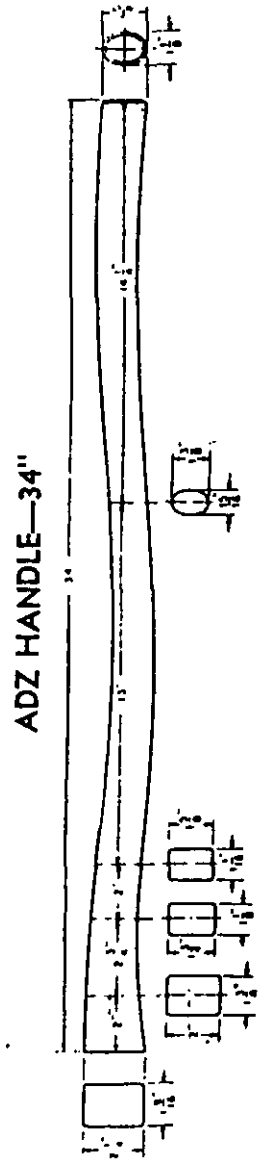
DOUBLE-BITTED AX HANDLE—36"

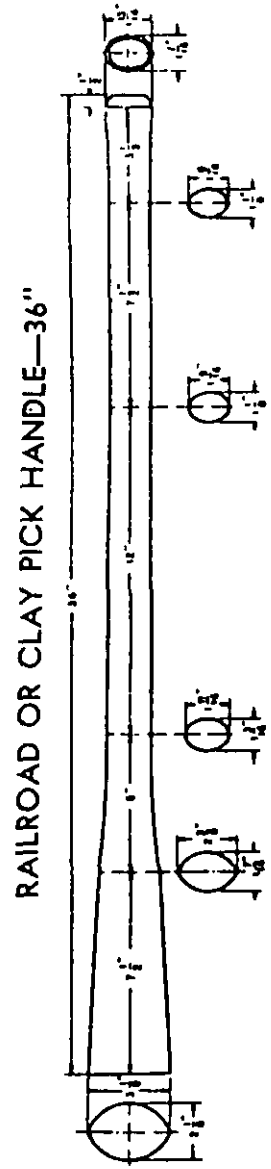


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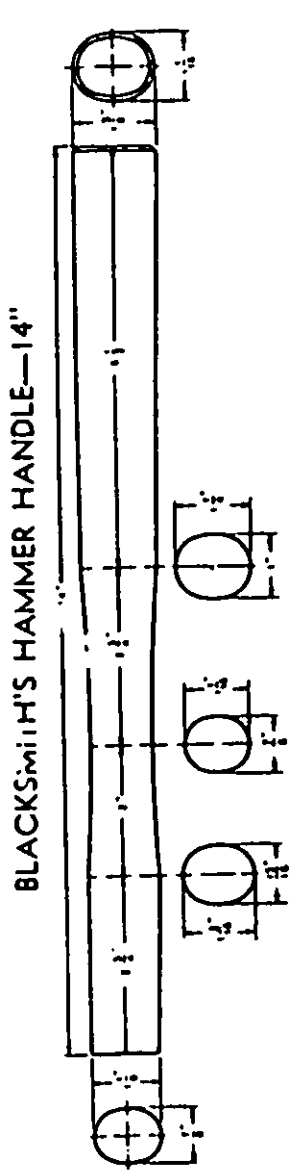


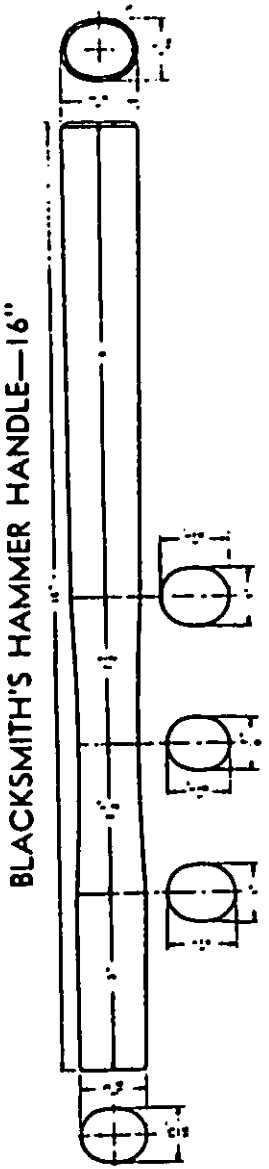


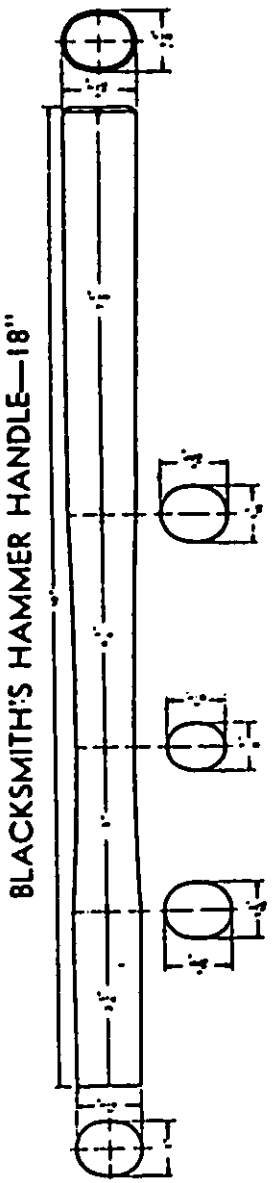


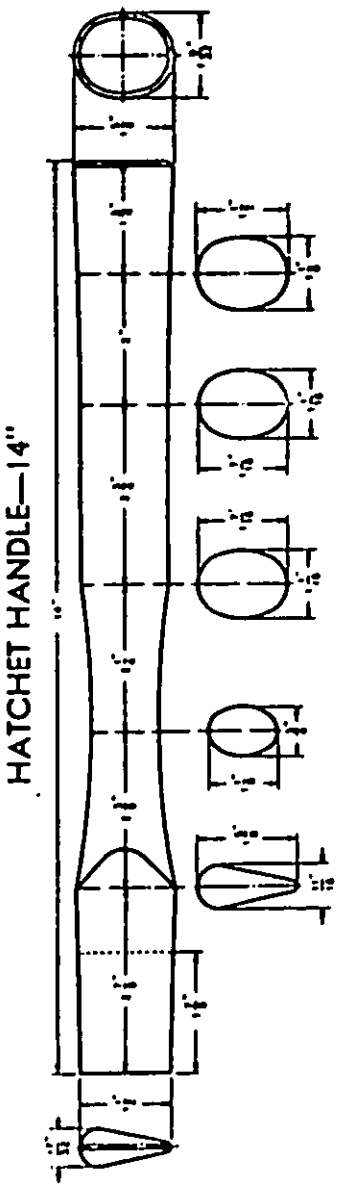


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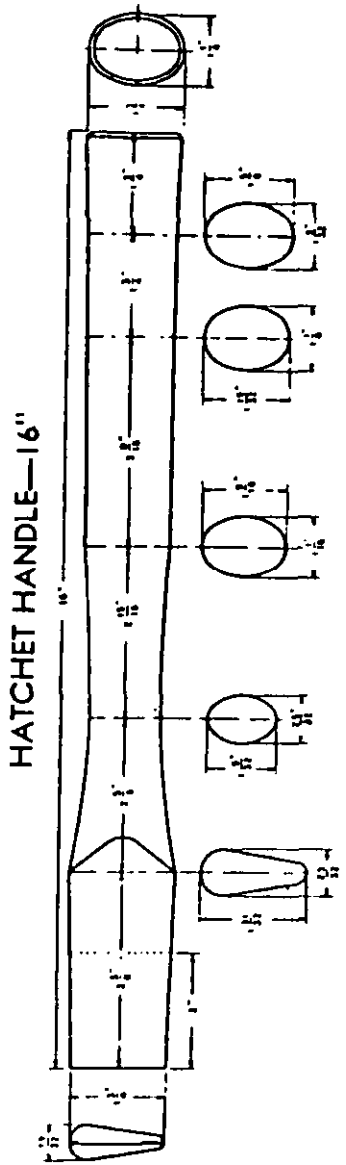


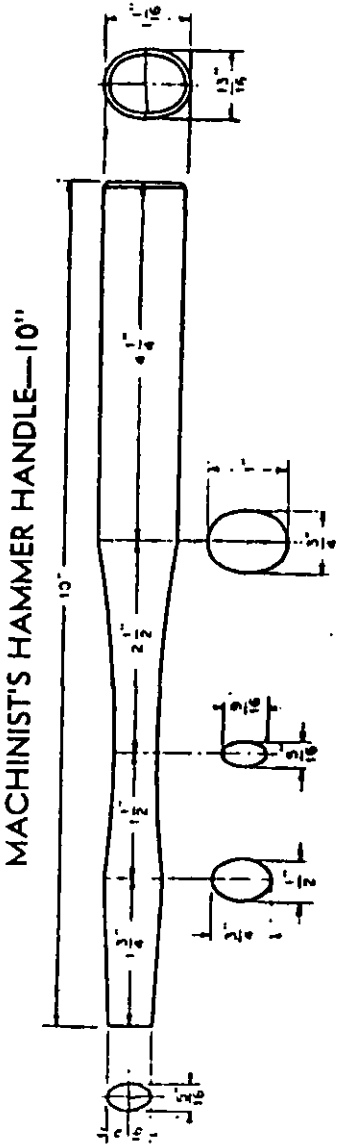


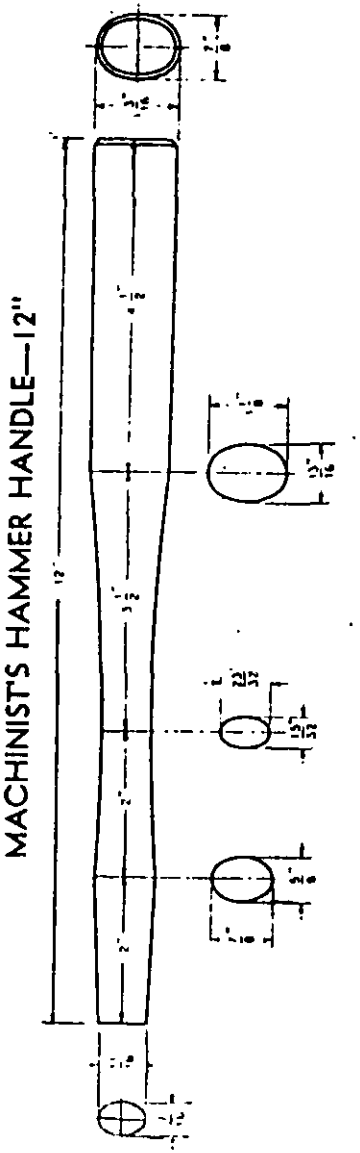


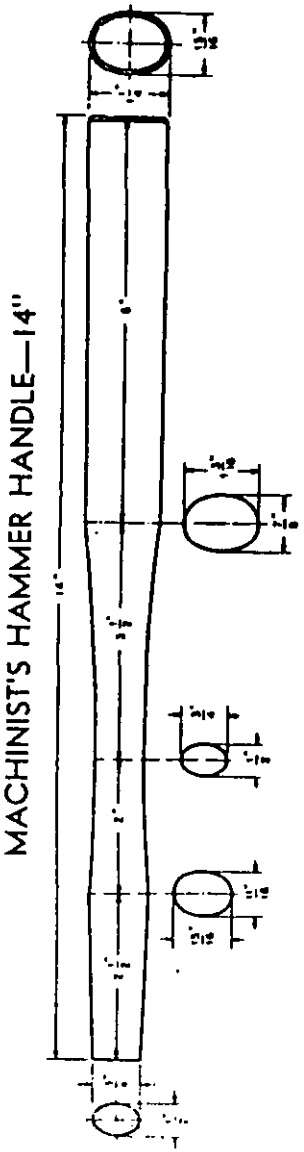


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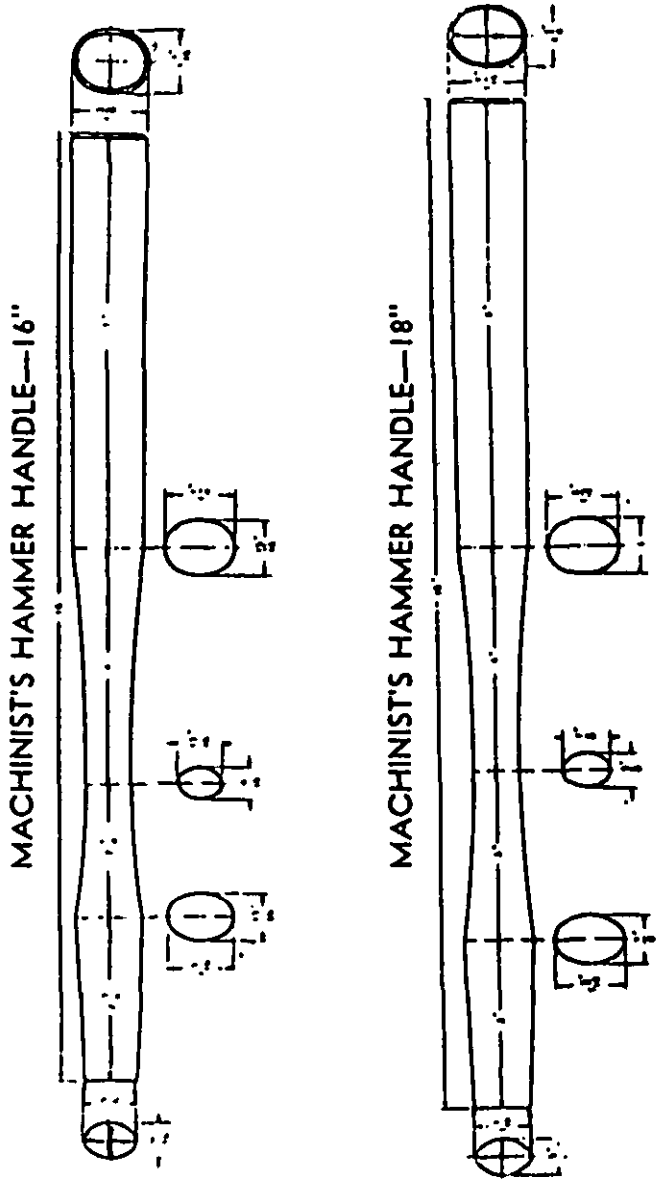








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E–1b. *Driven handles.*—Handles driven in tool heads by the manufacturer of the heads need not conform to the drawings to the same degree as handles without tool heads so long as they are suitable for use in the tool as determined by the inspector.

E–2. *Grades.*—Handles shall conform to the requirements for rate of growth, density, and maximum admissible defects and blemishes shown in table I.

TABLE I.—*Grades of hickory handles.*

Grade	Annual rings (number per inch of radius)	Weight [1] (pounds per cubic foot)	Maximum admissible defects and blemishes
AA	Not over 17	At least 55	Two small streaks [2] or equivalent in shorter streaks.[2]
A	Not over 23	At least 48	Four medium streaks [3] or equivalent in shorter streaks, light stain [4]

[1] At a moisture content of not over 12 percent.

[2] *Small streaks.*—Threadlike discolored lines extending not more than one-third the length of the handle.

[3] *Medium streaks.*—Discolored lines not over 1/32 inch wide extending more than one-third the length of the handle.

[4] *Light stain.*—Slight discoloration which does not materially impair the appearance of the handle.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—At least one handle shall be taken from each package, or one handle of each 24 shall be taken from packages containing more than 24 handles, for inspection and tests.

F–2. *Inspection and tests.*—Handles shall be inspected visually. Ordinarily it will not be necessary to weigh, count the annual rings, or determine the moisture content of each handle inspected. However, such detailed inspection shall be made in all cases where there is doubt as to the conformance of the handles inspected to the specification. Sample handles which do not conform to the specification shall be rejected. Lots from which rejected samples were taken shall be rejected.

F–2a. *Moisture content.*—When it is necessary to determine the moisture content of handles, either of the following methods shall be used.

F–2a(1). *Oven-drying method.*—Cut a section 1 inch long about 2 inches from the eye end of the handle. Weigh immediately on a scale that is accurate to ½ percent. This is the orginal weight. Dry in an oven maintained at a temperature of 212° to 221°F. (100° to 105°C.) until constant weight is attained. Reweigh the sample. This is the oven-dry weight. Compute moisture by the following formula:

$$\frac{\text{Original weight—oven-dry weight}}{\text{Oven-dry weight}} \times 100 = \text{Moisture content in percent.}$$

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F-2a(2). *Electrical moisture meter method.*—Determine the moisture content in accordance with instructions furnished by manufacturers of the meter. Electrical moisture meters shall not be used on handles that are surface wet or that have been color-stained.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G-1. *Packaging.*—Unless otherwise specified, commercial packages are acceptable.

G-2. *Packing.*—The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

G-3. *Marking.*—Shipping containers shall be marked with the name of the article, the size, type, grade, and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H-1. The following departmental specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, shall form a part of this specification, and shall be applicable to purchases made under this specification by the respective departments.

H-2. *Army.*—

H-2a. *Applicable specifications.*—

Joint Army-Navy Specifications:

JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN-P-128—Packaging and Packing for Overseas Shipment—Tape; Water Resistant, Gummed.
JAN-P-130—Packaging and Packing for Overseas Shipment—Barrier-Material; Utility, Noncorrosive, Paper.

U. S. Army Specification:

100-2—Standard Specification for Marking Shipments by Contractors.

H-2b. *Packaging.*—

H-2b(1). *For domestic shipment.*—Unless otherwise specified in the contract or order, commercial packages will be acceptable.

H-2b(2). *For overseas shipment.*—Handles with a length of 24 inches or less, of the same size and length, shall be bundled six to the bundle and securely tied at each end. Handles over 24 inches, of the same size and length, shall be bundled two to the bundle and securely tied at each end. Each individual bundle shall be wrapped with type I utility barrier-material conforming to the requirements of Joint Army-Navy Specification JAN-P-130, secured by wrapping with gummed tape conforming to the requirements of Joint Army-Navy Specification JAN-P-128, width suitable for the particular application.

H-2c. *Packing.*—

H-2c(1). *For domestic shipment.*—Packing shall conform to the requirements of paragraph G-1.

H-2c(2). *For overseas shipment.*—Bundles of handles of the same size and length shall be packed in snug-fitting nailed wood boxes conforming to the requirements of Joint Army-Navy Specification JAN-P-106, style 2, 2½, 3, or 4, for type 3 load, in quantities not to exceed 70 pounds. Boxes shall be closed and strapped in accordance with requirements of the appendix of the box specification.

H-2d. *Marking for shipment.*—Marking for shipment shall conform to the applicable requirements of U. S. Army Specification No. 100-2.

H-3. *Navy.*—

H-3a. *Applicable specifications and other publications.*—

Joint Army-Navy Specifications:

JAN-P-105—Packaging and Packing for Overseas Shipment—Boxes; Wood, Cleated, Plywood.
JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN-B-107—Boxes, Wood, Wirebound (Overseas Type).

Navy Department Specification:

General Specifications for Inspection of Material.

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.

Copies of Federal and Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. When requesting specifications, refer to both title and number (or symbol).

H-3b. *Packing.*—

H-3b(1). *For domestic shipment*[1].—Unless otherwise specified, handles shall be packed in either corrugated or solid fiberboard full-overlap end-opening boxes conforming to the following requirements.

[1] Packing requirements for domestic shipment apply only for direct Navy purchases. Unless otherwise specified, packing requirements for domestic shipments to subcontractors, contractors, or shipbuilders may be in accordance with best commercial practice.

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Gross weight, maximum	Average bursting strength, minimum	Sum of inside dimensions, maximum
Pounds	*Pounds*	*Inches*
40	200	60
65	275	75
90	350	90

Each box shall have a top and bottom pad equal in strength to the body material. The corner joint shall be metal-stitched. All flaps shall be sealed by means of a suitable adhesive, metal-stitching or tape. If taped, kraft gummed tape of not less than 2½-inch width, 60-pound minimum basis weight shall be used. Each box shall be securely strapped with not less than one lengthwise and one girthwise formed-edge flat metal strap, ⅝ by 0.020-inch minimum dimensions.

H-3b(2). *For overseas shipment.*—Handles, packed as described in paragraph H-3b(1), shall be overpacked in cleated plywood boxes, nailed wood boxes, or wire-bound wood boxes conforming to the requirements of Joint Army-Navy Specifications JAN-P-105, JAN P 106, and JAN-B-107, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds.

H-3b(3). *For domestic or overseas shipment.*—Where practicable, shipping containers shall be the same uniform size, and shall contain the identical number of items of one type, grade, and size. Containers shall be designed to fit the contents in a compact manner.

H-3c. *Marking.*—In addition to any special marking required by the contract or order, marking of shipping containers shall conform to the requirements of the Navy Shipment Marking Handbook.

H-3d. *Ordering information.*—Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed and shipping containers marked for domestic or overseas shipment. (See para. H-3b and H-3c.)

NAVY: SM MC O S Y

H-4. *Marine Corps.*—

H-4a. *Packing for overseas shipment.*—Unless otherwise specified, the handles shall be packed as described in paragraph H-3b(2).

H-4b. *Marking for overseas shipment.*—In addition to any special marking required by the contract or order, shipping containers shall be legibly marked with waterproof ink as follows:

Quantity
Name of article
Size
Type
Grade
Stock number
Contractor's name
Contract number and date

I. NOTES

I-1. Purchasers should exercise any desired options offered herein and should specify type, size, and grade required. (See para. B-1, B-1a, B-2, G-1, H-2b(1), H-3b(1), and H-4a.)

I-2. The technical requirements of this specification except for omission of requirements as to natural color of the wood, are practically the same as those contained in Simplified Practice Recommendation No. R77-45, Handles, Hickory, promulgated by the National Bureau of Standards November 1, 1945, and in use by the hickory-handle industry.

I-3. Strength tests conducted by the Forest Products Laboratory of the Forest Service, United States Department of Agriculture, at Madison, Wis., have shown that, weight for weight, sound straight-grained seasoned hickory has the same strength, toughness, and resistance to shock, regardless of whether it is heartwood (red hickory), sapwood (white hickory), or mixed hickory (heartwood and sapwood). A reliable indication of the strength of hickory is its density. Of two pieces of the same size and dryness, the heavier will be found to have the better strength properties. Rapid-growth hickory is generally stronger than slow-growth hickory, so that a guide to strength is rate of growth as determined from the rings of annual growth per inch of radius. Few growth rings per inch (fast growth), as shown on the end of a handle, indicates a stronger and tougher handle than many rings (slow growth), provided, of course, that it is straight-grained and free from weakening defects.

I-4. Defects and blemishes which may occur in hickory handles but which are not permissible under this specification, are as follows:

Birdpeck—Small break in the fiber of the wood, usually accompanied by a large streak.

Decay—Deterioration due to the action of wood-destroying fungi.

Failure—Area on a handle that failed to turn due to lack of wood.

Slight failure (see failure)—Limited to one-sixth of the contour of the handle if it does not affect the fit of the handle in the tool.

Cross grain—Deviation of the fiber from a line parallel to the axis of the handle in excess of 1 inch, in a length of 20 inches (1 in 20).

Abrupt dip grain—Local deviation of the fiber from its general direction in excess of ⅛ inch, in a distance of 4 inches.

Slight dip grain—Local deviation of the fiber from its general direction not in excess of ⅛ inch, in a distance of 4 inches.

Hole—May extend partially or entirely through the handle and may be from any cause.

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Knot—A portion of a branch or limb embedded in the wood. A sound, tight knot is solid across its face, as hard as the surrounding wood and so fixed by growth or position as to retain its place.

Mismanufacture—Imperfections produced in manufacture (includes failure).

Split—Lengthwise separation of the wood extending through the handle.

Heavy stain—Pronounced discoloration, predominantly blue, occurring in specks, spots, streaks or patches.

Large streaks—Discolored lines over 1/32 inch wide extending lengthwise.

I–5. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–6. Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

I–7. An Index of Federal Specifications may be purchased as noted in the paragraph below, price to be obtained from the Superintendent of Documents.

I–8. Copies of this specification may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price 10 cents.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

For sale by the Superintendent of Documents, Washington 25, D. C.
Price 10 cents.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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